

Fundamentals of Linear Control:

A Concise Approach

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Chapter 4: Feedback Analysis

linearcontrol.info/fundamentals

Recall from Chapter 3

Time-invariant linear system in the time-domain

Impulse response

$$\begin{aligned} g(t), \quad t \geq 0 \end{aligned}$$

Convolution

$$\begin{aligned} y(t) &= \int_{0^-}^{\infty} g(t - \tau) u(\tau) d\tau \end{aligned}$$

Time-invariant linear system in the frequency-domain

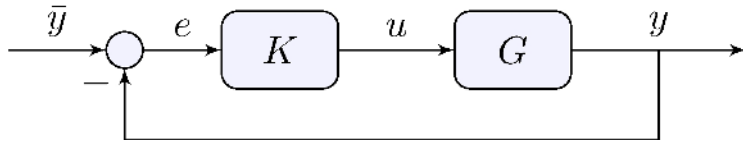
Transfer-function

$$\begin{aligned} G(s) = \mathcal{L}\{g(t)\} \end{aligned}$$

Product

$$\begin{aligned} Y(s) &= G(s) U(s), & Y(s) &= \mathcal{L}\{y(t)\}, & U(s) &= \mathcal{L}\{u(t)\} \end{aligned}$$

4.1 Tracking, Sensitivity, and Integral Control



Loop relations

$$\begin{aligned} Y(s) &= G(s) \cdot U(s), & U(s) &= K(s) \cdot E(s), & E(s) &= \bar{Y}(s) - Y(s) \end{aligned}$$

Closed-loop sensitivity transfer-function

$$\begin{aligned} E(s) &= S(s) \cdot \bar{Y}(s), & S(s) &= \frac{1}{1 + G(s) \cdot K(s)} \end{aligned}$$

Short-hand notation

$$\begin{aligned} e &= S \cdot \bar{y}, & S &= \frac{1}{1 + G \cdot K}, & \text{etc} \end{aligned}$$

Example: car model with proportional feedback

Open-loop transfer-function

$$\begin{aligned} G(s) = \frac{\frac{p}{m}}{s + \frac{b}{m}} \end{aligned}$$

Closed-loop sensitivity transfer-functions

$$\begin{aligned} S(s) = \frac{1}{1 + K \frac{\frac{p}{m}}{s + \frac{b}{m}}} = \frac{s + \frac{b}{m}}{s + \frac{b}{m} + K \frac{p}{m}} \end{aligned}$$

Closed-loop stability

$$\begin{aligned} - \left(\frac{b}{m} + \frac{p}{m} K \right) < 0 \quad \&\implies \quad K > - \frac{b}{p} \end{aligned}$$

Tracking error

Reference input

$$\backslash[\backslashbegin{align*}\bar{y}(t) \&= \bar{y} \cos(\omega t + \phi), \quad t \geq 0 \end{align*}\backslash]$$

Steady-state from frequency response

$$\backslash[\backslashbegin{align*} e_{\mathrm{ss}}(t) \&= \bar{y}, |S(j\omega)| \cos(\omega t + \phi + \angle S(j\omega)) \& \& \implies |e_{\mathrm{ss}}(t)| \&\leq |S(j\omega)|, |\bar{y}| \end{align*}\backslash]$$

e.g. car model with a step reference

$$\backslash[\backslashbegin{align*} |e_{\mathrm{ss}}(t)| \&\leq |S(0)|, |\bar{y}| = \frac{\frac{b}{m}}{\frac{b}{m} + \frac{p}{m} K} \& \& |\bar{y}| = \frac{1}{1 + \frac{p}{b} K} |\bar{y}| \end{align*}\backslash]$$

Asymptotic tracking

Definition

Error converges to zero for large time $\lim_{t \rightarrow \infty} e(t) = 0$

Stability of (S) is not enough. It is necessary that $E = S^{-1} \bar{Y}$ Poles of $\bar{Y}$ must be cancelled by zeros of (S) !

Lemma 4.1

Let $(S = (1 + G, K)^{-1})$ be asymptotically stable and $\bar{y}(t) = \bar{y} \cos(\omega t + \phi)$
If (G, K) has a pole at $(s = j\omega)$ then $\lim_{t \rightarrow \infty} e(t) = 0$

Proof: (S) has a zero at $(s = j\omega)$

Example: toilet bowl model

Open-loop transfer-function

$$G(s) = \frac{1}{sA}$$

Closed-loop sensitivity transfer-function

$$S(s) = \frac{1}{1 + G(s)K} = \frac{1}{1 + \frac{K}{sA}} = \frac{s}{s + \frac{K}{A}}$$

Asymptotically stable for $(K > 0)$

$S(s)$ has a zero at zero: asymptotic tracking of step (constant) reference inputs

Example: car model with integral feedback

Open-loop transfer-function

$$\begin{aligned} G(s) = \frac{\frac{p}{m}}{s + \frac{b}{m}} \end{aligned}$$

Integral control

$$\begin{aligned} K(s) = \frac{K}{s} \end{aligned}$$

Closed-loop sensitivity transfer-functions

$$\begin{aligned} S(s) = \frac{1}{1 + \frac{K}{s} \frac{\frac{p}{m}}{s + \frac{b}{m}}} = \frac{s \left(s + \frac{b}{m} \right)}{s^2 + \frac{b}{m} s + K \frac{p}{m}} \end{aligned}$$

Asymptotically stable for $(K > 0)$

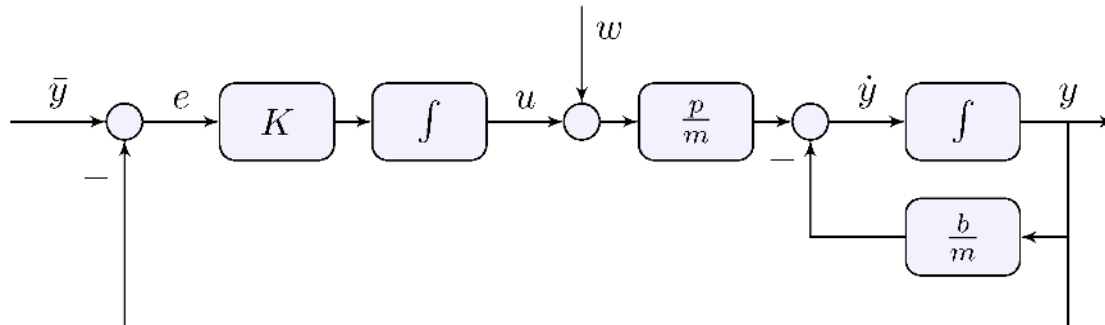
(S) has a zero at zero: asymptotic tracking of step (constant) reference inputs

Example: car model with integral feedback

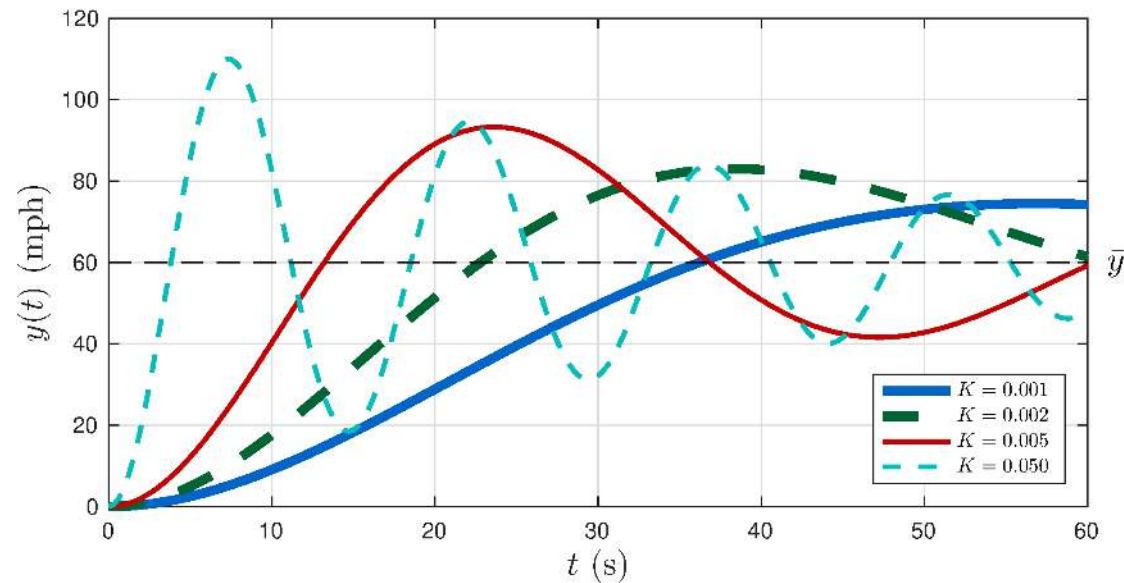
Controller

$$U(s) = \frac{K}{s} E(s) \implies u(t) = u(0) + K \int_0^t e(\tau) d\tau$$

System + Controller block-diagram



Example: car model with integral feedback



Transient response has oscillations!

4.2 Stability and Transient Response

Some facts

- The zeros of $(1 + G K)$ are the poles of (S)
- The poles of the product (G, K) are the zeros of (S)
- When (G) and (K) are rational the order of (S) is the order of the product (G, K)

Example: rational transfer-function with no pole-zero cancellations

$$\begin{aligned} G &= \frac{N_G}{D_G}, & K &= \frac{N_K}{D_K}, & S &= \frac{1}{1 + \frac{N_G}{D_G} \frac{N_K}{D_K}} \\ & & & & & \frac{D_G D_K}{D_G D_K + N_G N_K} \end{aligned}$$

For tracking

- Add poles to (K) so they become zeros of (S)
- More complex (K) leads to more complex (S)
- Watch out for stability and oscillations, e.g. car model with integral feedback

Example: car model with proportional plus integral feedback

Proportional plus integral control

$$\begin{aligned} G(s) &= \frac{p}{s + b}, & K(s) &= K_p + s^{-1} K_i \end{aligned}$$

Closed-loop sensitivity transfer-functions

$$\begin{aligned} S(s) &= \frac{1}{1 + G(s)K(s)} = \frac{1}{1 + \frac{p}{m} (K_p s + K_i) \left(s + \frac{b}{m} \right)} = \frac{s \left(s + \frac{b}{m} \right)}{s^2 + \left(\frac{b}{m} + \frac{p}{m} K_p \right) s + \frac{p}{m} K_i} \end{aligned}$$

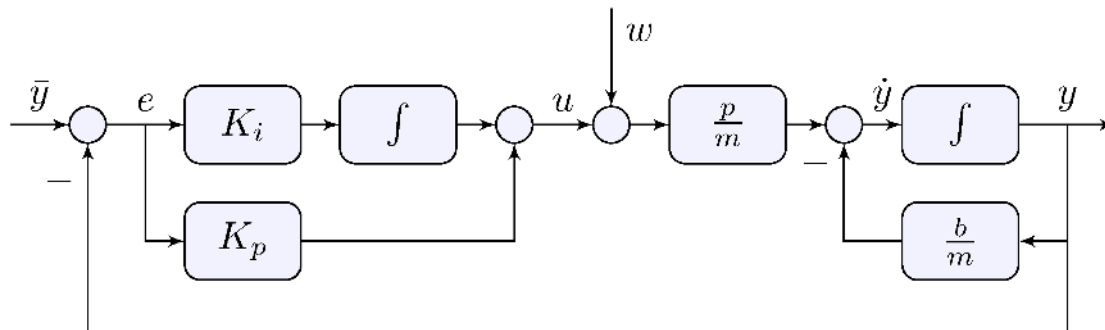
$$\text{No complex-poles (oscillations) if } 0 < 4 \frac{p}{m} K_i \leq \left(\frac{b}{m} + \frac{p}{m} K_p \right)^2$$

Example: car model with proportional plus integral feedback

Controller

$$\begin{aligned} U(s) &= \left(K_p + \frac{K_i}{s} \right) E(s) \implies u(t) = u(0) + K_p e(t) + K_i \int_0^t e(\tau) d\tau \end{aligned}$$

System + Controller block-diagram



Example: car model with proportional plus integral feedback

Special choice

$$\begin{aligned} K_i &= \frac{b}{m} K_p, & K(s) &= K_p + \frac{K_i}{s} = K_p \frac{s + \frac{b}{m}}{s} \end{aligned}$$

leads to a pole-zero cancellation
$$\begin{aligned} G(s), K(s) &= \frac{\frac{p}{m}}{s + \frac{b}{m}} \times K_p \\ \frac{s + \frac{b}{m}}{s} &= \frac{\frac{p}{m} K_p}{s}, \end{aligned}$$

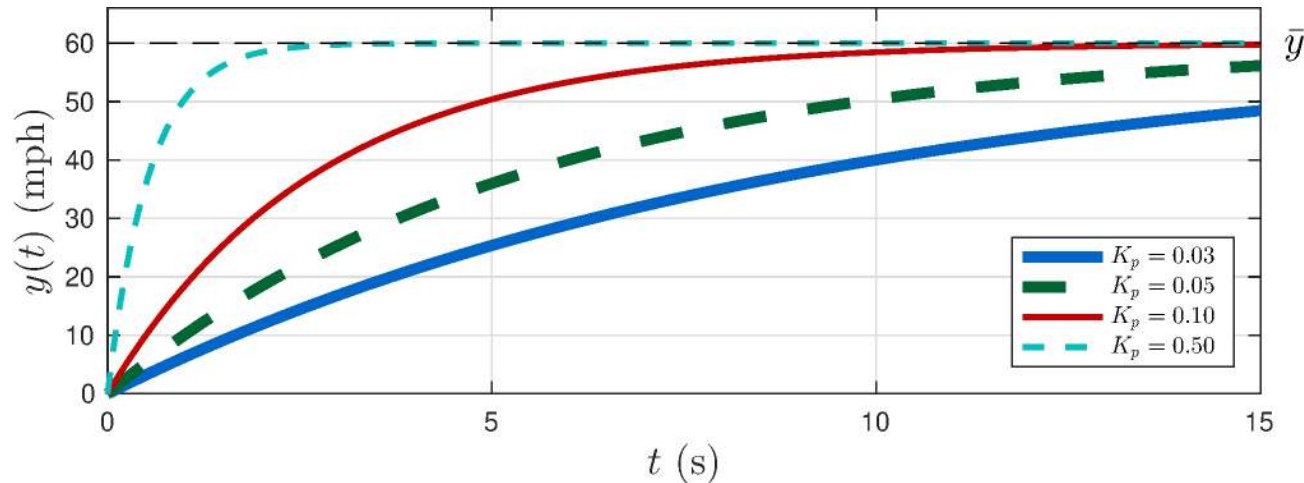
Closed-loop sensitivity transfer-functions

$$\begin{aligned} \label{eq:scarpipzc} S(s) &= \frac{1}{1 + G(s) K(s)} = \frac{s}{s + \frac{p}{m} K_p} \end{aligned}$$

Zero at the origin: asymptotic tracking

No complex poles: no oscillations

Example: car model with proportional plus integral feedback



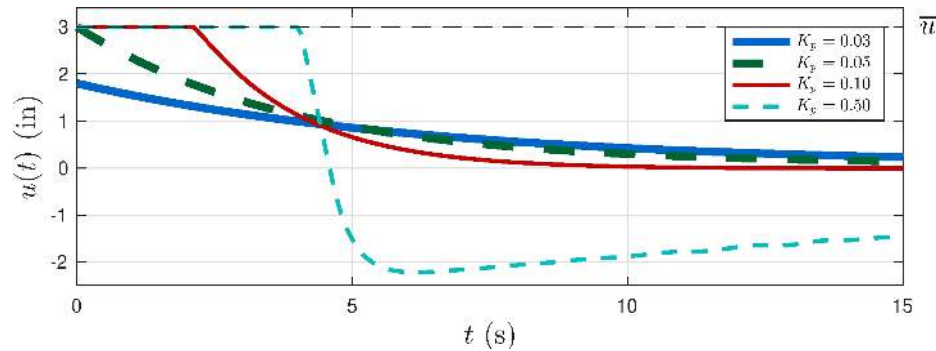
Asymptotic tracking, gone are the oscillations!

Beware of control effort: saturation occurs if $K_p > \frac{\overline{u}}{\bar{y}} = \frac{3}{60} = 0.05$

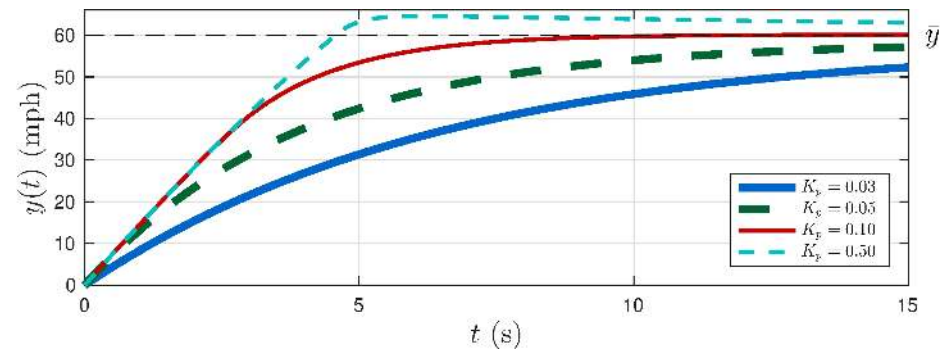
4.3 Integrator Wind-up

Example: car model with proportional plus integral feedback

Nonlinear input saturation

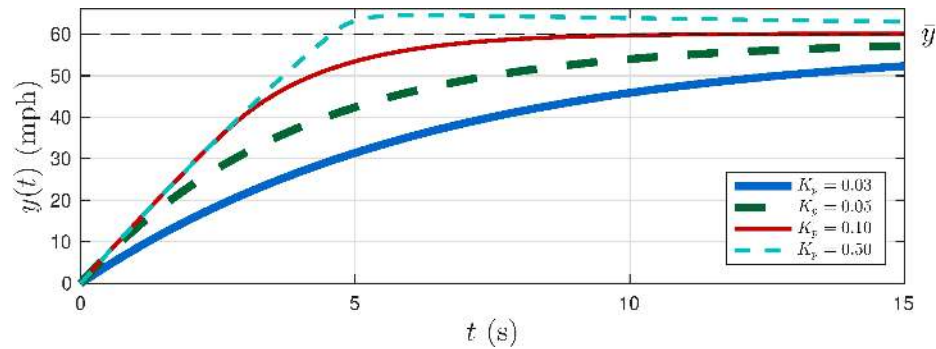


Output overshoots

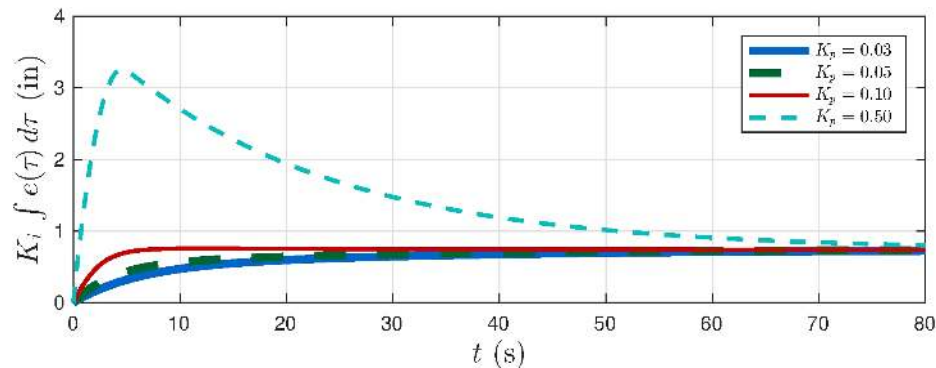


Example: car model with proportional plus integral feedback

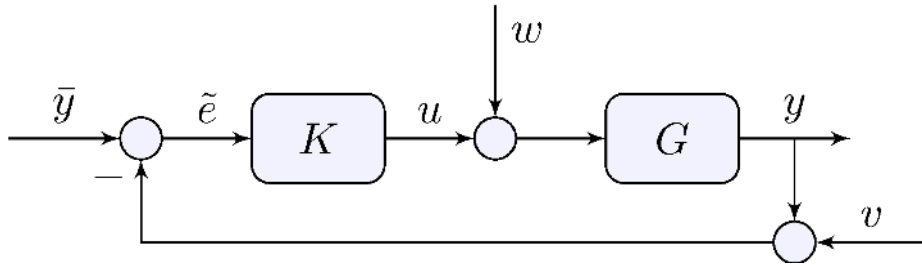
Output converges slowly



Integrator wind-up



4.4 Feedback with Disturbances



Loop relations

$$\begin{aligned} y &= G(u + w), & u &= K\tilde{e}, & \tilde{e} &= \bar{y} - (y + v) \end{aligned}$$

Closed-loop transfer-functions

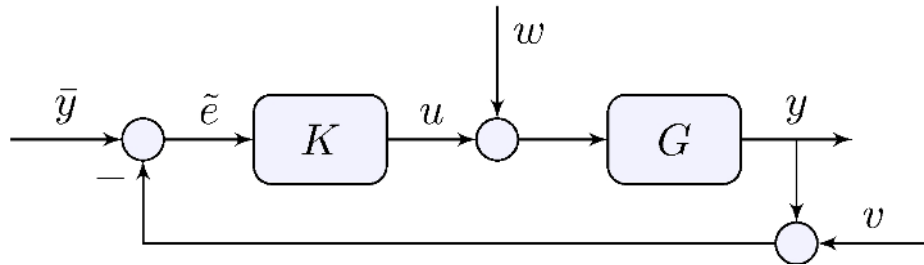
$$\begin{aligned} y &= H(\bar{y} - H(y + v) + D(w)), & u &= Q(\bar{y} - Q(y + v) - H(w)) \end{aligned}$$

Tracking error

$$e = \bar{y} - y = S(\bar{y} + H(y + v) - D(w))$$

Beware $\tilde{e} \neq e$!

4.4 Feedback with Disturbances



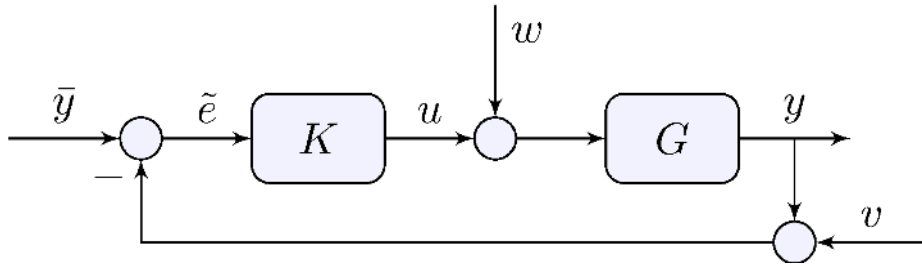
Closed-loop transfer-functions

$$\begin{aligned} y &= H(\bar{y} - H(y - Qv + Dw)) \\ u &= Q(\bar{y} - Qv - H(y - Dw)) \\ e &= S(\bar{y} + H(y - Dw) - v) \end{aligned}$$

Gang of four

$$\begin{aligned} S &= \frac{1}{1 + GK}, & D &= \frac{G}{1 + GK} & H &= \frac{GK}{1 + GK}, & Q &= \frac{K}{1 + GK} \end{aligned}$$

4.4 Feedback with Disturbances

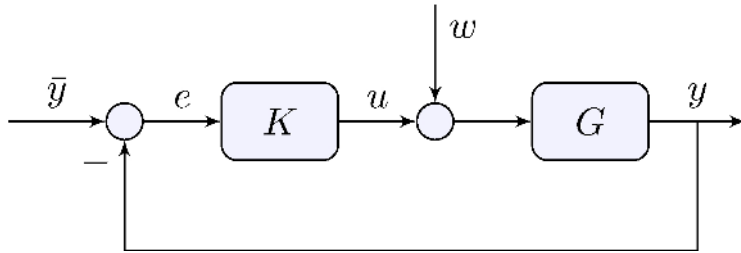


Gang of four

$$\begin{aligned} S &= \frac{1}{1 + G K}, & D &= G S, & H &= G K S, & Q &= K S \end{aligned}$$

- S has as zeros the poles of $G K$
- No pole-zero cancellations in $G K$ $\implies S, D, H, Q$ have the same poles!
- Stabilizing S stabilizes all other transfer-functions
- If there are pole-zero cancellations in $G K$ be careful and see Section 4.7

4.5 Input-Disturbance Rejection



Closed-loop transfer-functions

$$\begin{aligned} e &= \bar{y} - D \cdot w, & D &= G S = \frac{G}{1 + G K} \end{aligned}$$

Lemma 4.2

Let $(S = (1 + G K)^{-1})$ and $(D = G S)$ be asymptotically stable and $\begin{aligned} \bar{y}(t) &= \bar{y} \cos(\omega t + \phi), \\ w(t) &= \bar{w} \cos(\omega t + \psi) \end{aligned}$ If (K) has a pole at $(s = j \omega)$ then $(\lim_{t \rightarrow \infty} e(t) = 0)$

Proof: (S) and (D) have zeros at $(s = j \omega)$

Example: car model with proportional feedback

Closed-loop transfer-functions

$$\begin{aligned} S(s) &= \frac{s + \frac{b}{m}}{s + \frac{b}{m} + \frac{p}{m} K}, & D(s) &= S(s), & G(s) &= \frac{\frac{p}{m}}{s + \frac{b}{m} + \frac{p}{m} K} \end{aligned}$$

No asymptotic tracking, no asymptotic disturbance rejection

Example: car model with integral feedback

Closed-loop transfer-functions

$$\begin{aligned} S(s) &= \frac{s \left(s + \frac{b}{m} \right)}{s^2 + \frac{b}{m} s + \frac{p}{m} K}, & D(s) &= S(s), & G(s) &= \frac{\frac{p}{m} s}{s^2 + \frac{b}{m} s + \frac{p}{m} K} \end{aligned}$$

Asymptotic tracking and asymptotic disturbance rejection, oscillations

Example: car model with integral feedback

Closed-loop transfer-functions

$$\begin{aligned} S(s) &= \frac{s \left(s + \frac{b}{m} \right)}{s^2 + \frac{b}{m} s + \frac{p}{m} K}, & D(s) &= S(s) \cdot, \\ G(s) &= \frac{\frac{p}{m} s}{s^2 + \frac{b}{m} s + \frac{p}{m} K} \end{aligned}$$

Asymptotic tracking and asymptotic disturbance rejection, oscillations

Example: car model with proportional plus integral feedback

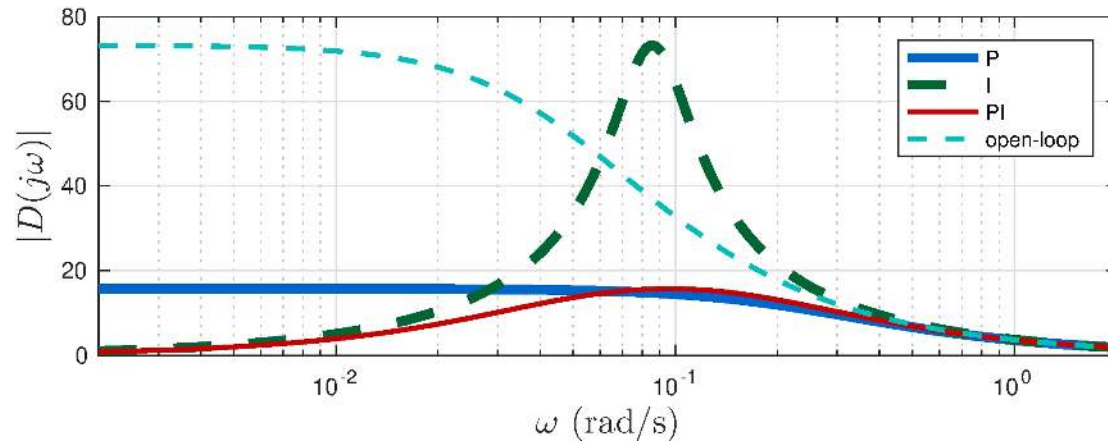
Closed-loop transfer-functions

$$\begin{aligned} S(s) &= \frac{s}{s + \frac{p}{m} K_p}, & D(s) &= S(s) \cdot, \\ G(s) &= \frac{\frac{p}{m} s}{\left(s + \frac{b}{m} \right) \left(s + \frac{p}{m} K_p \right)} \end{aligned}$$

Asymptotic tracking and asymptotic disturbance rejection, no oscillations

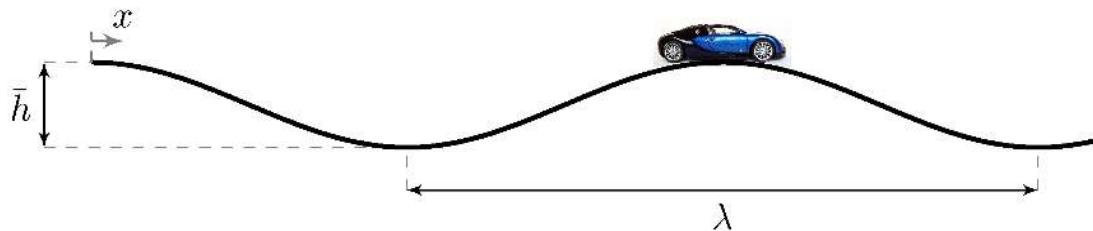
4.5 Input-Disturbance Rejection

Example: car model with various feedback controllers



Integral controller is sensitive to disturbances at $\omega = 0.1$ rad/s

Watch out for rolling hills!



Example: toilet bowl model

Loop relations

$$\begin{aligned} G(s) &= \frac{1}{sA}, & K(s) &= K \end{aligned}$$

Closed-loop transfer-functions

$$\begin{aligned} S(s) &= \frac{s}{s + \frac{K}{A}}, & D(s) &= \frac{\frac{1}{A}}{s + \frac{K}{A}} \end{aligned}$$

Asymptotic tracking but no asymptotic disturbance rejection!

Pole must be at the controller for asymptotic disturbance rejection

$D(s)$ does not have a zero at the origin

4.6 Measurement noise

Closed-loop transfer-functions

$$\begin{aligned} e &= S \bar{y} + H v, & S &= \frac{1}{1 + G K}, & H &= G K S \end{aligned}$$

Fundamental limitation

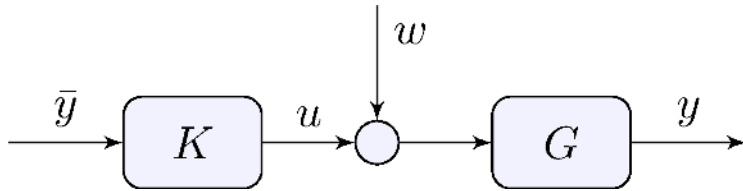
$$S + H = 1$$

At every frequency

$$S(j\omega) + H(j\omega) = 1$$

- Good tracking ($|S(j\omega)| \approx 0$) means bad measurement noise rejection ($|H(j\omega)| \approx 1$)
- Good measurement noise rejection ($|H(j\omega)| \approx 0$) means bad tracking ($|S(j\omega)| \approx 1$)
- Tracking requires good sensors!
- $|S(j\omega)| + |H(j\omega)| \neq 1$: both $|S|$ and $|H|$ can be large at the same time!

4.7 Pole-zero Cancellations and Stability



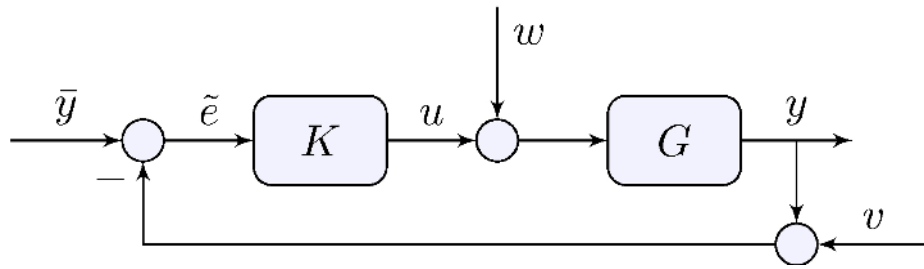
Open-loop

$$\begin{aligned} G &= \frac{N_G}{D_G}, \text{ \& } K = G^{-1} = \frac{D_G}{N_G}, \end{aligned}$$

- G cannot be strictly proper or K will not be proper
- Even if G is proper but not strictly proper and $K = G^{-1}$

$$y = G^{-1} K \bar{y} + G w = \bar{y} + G w, \text{ \& } u = K \bar{y}$$
- Both G and K must be stable or y or u will be unbounded
- Stabilization of unstable systems cannot be done in open-loop

4.7 Pole-zero Cancellations and Stability



Closed-loop

Internal stability if

S, G, K are all asymptotically stable

Lemma 4.3:

Internal stability if and only if (S) is asymptotically stable and any pole-zero cancellation in $(G K)$ is of stable poles or zeros

Proof (sketch): If
$$\begin{aligned} G &= \frac{(s - z)}{(s - p)} \tilde{G}, & K &= \frac{(s - p)}{(s - z)} \tilde{K}, & G K &= \tilde{G} \tilde{K}, \\ S &= \tilde{S} = \frac{1}{1 + \tilde{G} \tilde{K}} \end{aligned}$$
 However
$$\begin{aligned} S G &= \frac{(s - z)}{(s - p)} \tilde{S} \tilde{G}, & S K &= \frac{(s - p)}{(s - z)} \tilde{K} \tilde{G} \end{aligned}$$
 so that (z) and (p) must both have negative real part

Example: car model with proportional plus integral feedback

Model and controller

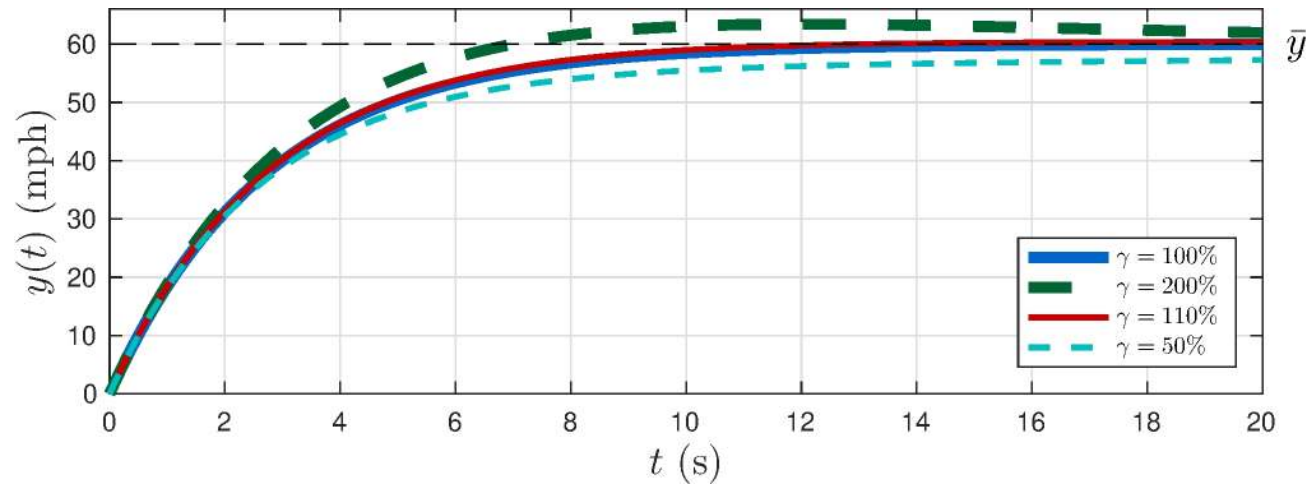
$$\begin{aligned} G(s) &= \frac{\left(\frac{p}{m} \right) (s + \left(\frac{b}{m} \right))}{s + z}, & K(s) &= K_p \\ & \frac{s + z}{s}, & z &= \frac{K_i}{K_p} = \frac{b}{m} \end{aligned}$$

Closed-loop transfer-functions

$$\begin{aligned} S(s) &= \frac{s}{s + \frac{p}{m} K_p}, & H(s) &= \frac{\frac{p}{m} K_p}{s + \frac{p}{m} K_p}, & Q(s) &= \frac{K_p}{s + \frac{p}{m} K_p} \\ & \frac{K_p}{s + \frac{p}{m} K_p}, & D(s) &= \frac{\frac{p}{m} s}{(s + \frac{b}{m}) (s + \frac{p}{m} K_p)} \end{aligned}$$

Canceled pole is a zero of $Q(s)$ and a pole of $D(s)$

Example: car model with proportional plus integral feedback



Imperfect cancellation

$$K_i = \gamma \frac{b}{m} K_p$$

More to come in Chapter 6

Learn more

- [Book website](#)

